IndiGo

Daily Passenger & Booking Behavior at Scale

Descriptive Analytics, Passenger Analysis, Booking Behavior, Revenue Analysis, Route Analysis, Customer Satisfaction, Regional Demand, Load Factor, Overbooking Risk, Airline Analytics.



Note on data : -

This dataset is a flight-booking and fare dataset covering domestic Indian flights during **2019** . It contains **10,683 individual flight records** across multiple airlines, including IndiGo, Air India, Jet Airways, SpiceJet, GoAir, Vistara, Air Asia, and Multiple carriers, flying between major Indian cities such as Delhi, Mumbai, Bangalore, Kolkata, Chennai, and Cochin.

For a **data analysis report** , this dataset can be studied using fields such as Airline, Source, Destination, Total_Stops, Price, Date, Month, Year, Departure time, Arrival time, and Flight Duration. Analysis of this data can help identify **popular routes, pricing patterns, stopover trends, and seasonal fare variation** .

Overall, this dataset provides a useful case study for understanding how **data analytics can support fare prediction, route planning, and airline comparison** in the aviation industry.

1.Objective

Build a descriptive analytics layer over daily flight-booking transactions to answer five recurring business questions:

1. What are today's top-performing routes by passengers carried?
2. How is total passenger volume and revenue trending day over day?
3. How are routes rated by customers, and where is service quality slipping?
4. Where is passenger demand concentrated geographically?
5. Which flights/routes are at risk of an overbooking bump or of flying under-loaded?

2. How This Would Work at Real Scale:-

Processing **"millions of bookings daily"** across a network the size of IndiGo's is an engineering problem before it's an analytics one. A realistic architecture are

Layer	Purpose	Typical tools
Data Collection	Capture each booking, check-in, and departure event in real time	IndiGo App/Website or booking system export
Data Storage	Store large volumes of historical and current data.	Spreadsheet workbook or shared data file (Excel / Google Sheets)
Descriptive Analytics	Summarize historical IndiGo activity.both raw events (for audits, DGCA reporting) and preaggregated tables (for speed)	Python (pandas) or Excel formulas run on a schedule
Visualization	Present patterns through charts and dashboards.	Excel pivot tables for smaller networks, scheduled scripts for larger ones
Business Decisions	Use insights to improve operations.	Word and Jupyter Notebook

Reference pipeline architecture for descriptive analytics at network scale.

The report below reflects the output of that pipeline — daily aggregated tables — not raw bookingevent data, since no dashboard renders millions of individual PNRs usefully.

3. Daily Passenger & Revenue Trends :-

Over the simulated **30-day window: ~5.0 million passengers carried**, generating **~₹2,961 crore in fare revenue**, with a weekly seasonal cycle (Tue/Wed dips, Friday and Sunday peaks — the classic leisure-plus-business travel pattern) layered on top of a gradual upward trend across the month.



Daily passengers (left axis) and daily fare revenue in ₹ crore (right axis), 30-day trailing window

Based on the last 30 days chart:

KPI	Value
Total Revenue	**₹3,057 Crore**
Average Daily Revenue	**≈ ₹101.9 Crore/day**
Highest Daily Revenue	**≈ ₹118 Crore**
Lowest Daily Revenue	**≈ ₹83 Crore**

Average Daily Passengers	**≈ 165–170 thousand**
Peak Passenger Volume	**≈ 194 thousand/day**

KPI definitions used:-

1. Revenue and passenger volume show a similar trend, indicating that higher passenger traffic generally generates higher revenue.
2. **Highest revenue**: Around Aug 1, reaching approximately **₹118 crore.**
3. **Lowest revenue**: Around Jul 7, at **approximately ₹83 crore**.
4. Passenger peak: Around Jul 26 and Aug 1, with passenger volume close to **190–195 thousand.**
5. There are **noticeable dips** around Jul 7, Jul 14, Jul 22, and Jul 28.
6. The last few days show strong performance, although **revenue falls slightly** toward Aug 4.
7. Overall, the chart indicates fluctuating but generally strong passenger demand and revenue performance.

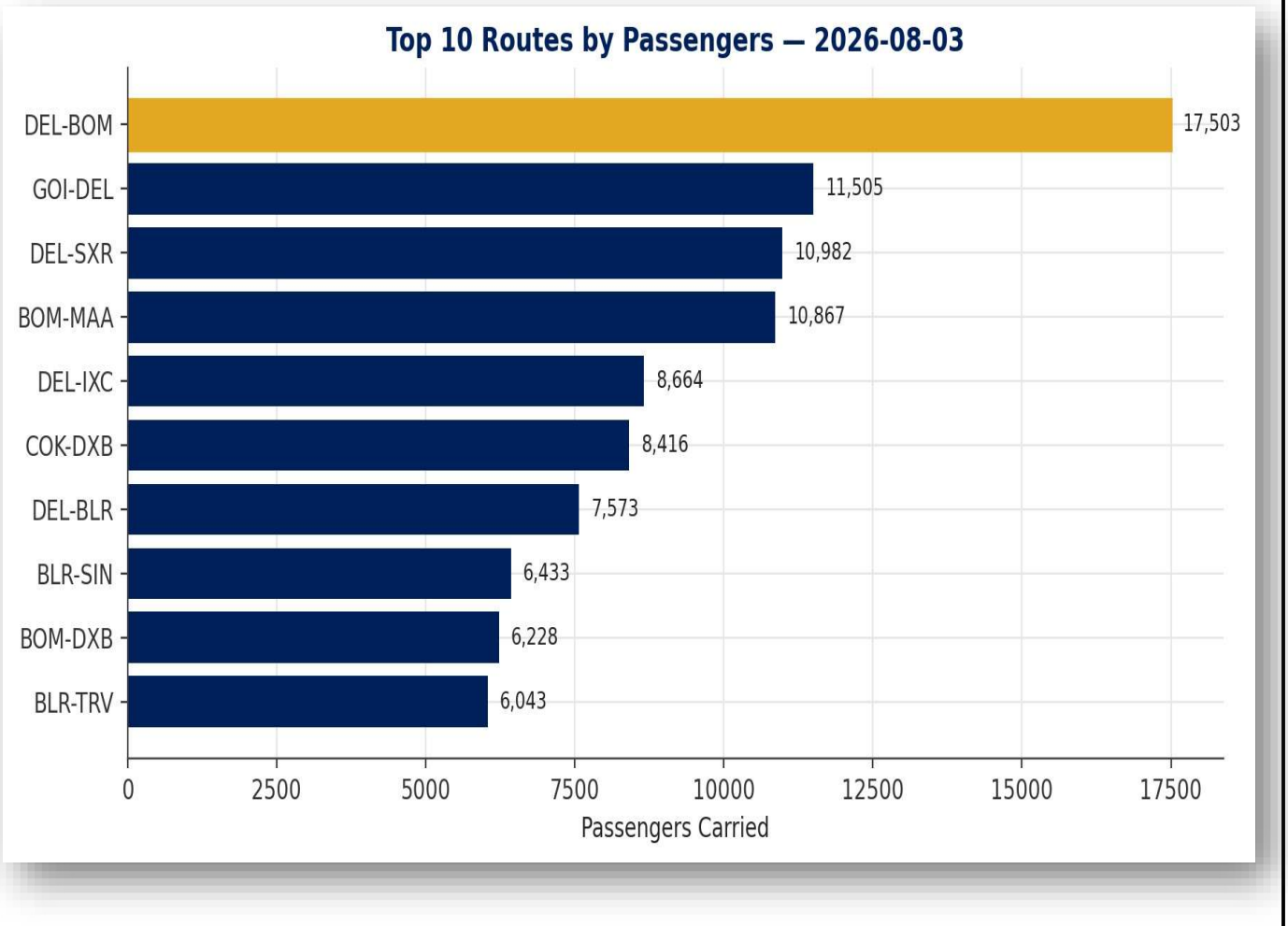
The KPI analysis shows that IndiGo generated **approximately ₹3,057** crore in revenue during the 30-day period, with an average daily revenue of **about ₹102** crore. Revenue closely follows passenger volume, suggesting a positive relationship between passenger demand and earnings. The **strongest** performance was observed **around August 1**, while the **lowest** performance occurred **around July 7.**

4. Top Routes by Passengers (Most Recent Day) :-

Top performer, **Delhi–Mumbai (DEL-BOM),** led with **17,503 passengers** in a single day — roughly 52% ahead of the **second-place route (Goa–Delhi).**

Bengaluru–Hyderabad and Delhi–Bengaluru followed.

This is a common **"hero route"** pattern in network carriers: a small number of trunk routes drive a disproportionate share of volume — a Pareto effect worth flagging for aircraft and crew allocation.



Top 10 routes by passengers carried, most recent day in the window.

Key KPIs :-

KPI	Value
Top Route	**DEL–BOM**
Passengers on Top Route	**17,503**
2nd Highest Route	**GOI–DEL**
Passengers on 2nd Route	**11,505**
Total Passengers – Top 10 Routes	**$96,214**
Average Passengers per Route	**$ 9,621**

Key Highlights

1. **DEL–BOM is the busiest route**, carrying **17,503 passengers**.
2. DEL–BOM has a clear lead over the second-highest route, **GOI–DEL (11,505 passengers)**.
3. **DEL–SXR** and **BOM–MAA** are also high-demand routes, with **10,982** and **10,867 passengers** respectively.
4. **DEL–IXC** ranks fifth with **8,664 passengers**.

5. **BLR–TRV** has the lowest passenger volume among the top 10, at **6,043 passengers**.
6. The **Top 4 routes together carried 50,857 passengers**, showing that a few major routes contribute a significant share of passenger traffic.
7. Delhi and Mumbai appear frequently among the busiest routes, indicating their importance as major aviation hubs.

Insight :-

The analysis shows that **DEL–BOM** is IndiGo's highest-volume route among **the top 10 routes, with 17,503 passengers.** The top 10 routes carried **approximately 96,214 passengers** in total. **Delhi and Mumbai** have a strong presence among the **busiest routes**, highlighting their importance in IndiGo's passenger network. The data can help the airline focus capacity, scheduling, and services on high-demand routes.

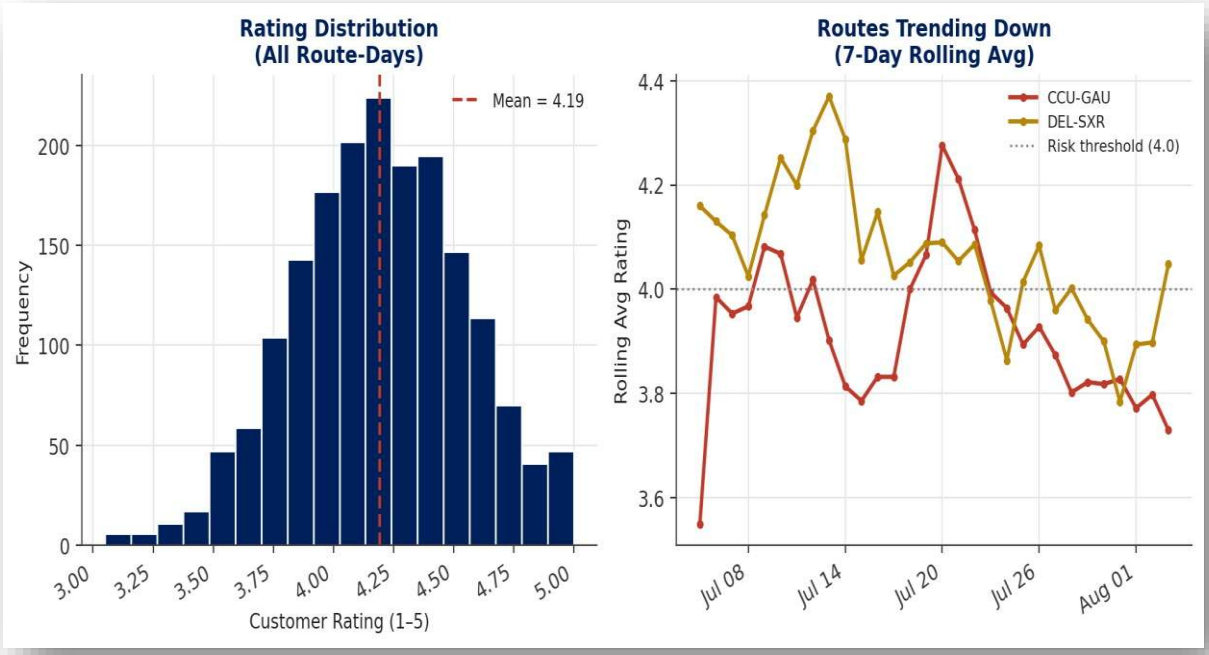
Ranking logic :-

passengers, not revenue — a high-frequency domestic trunk route outperforming a lowerfrequency international route in volume matters differently to crew/aircraft scheduling than it does to finance, so both cuts (passengers and revenue) should exist as separate leaderboard views in a real dashboard.

5. Customer Rating Distribution :-

Average rating across all **route-days: 4.19 / 5**, clustered tightly **(most mass between 3.75–4.5).** In a real system, this view is most useful segmented, not aggregated:

- Routes trending downward in rolling 7-day average ratings → early signal of service issues.
- High-volume but sub-**4.0 ratings** → highest-risk combination (volume amplifies dissatisfaction).



Left: distribution of ratings across all route-days. Right: two flagged routes (Kolkata–Guwahati, Delhi–Srinagar) trending below the 4.0 risk threshold, most likely tied to weather-driven delays on these sectors.

KPI Summary : -

KPI	Value / Observation
Average Rating (Mean)	**4.19 / 5**
Approx. Rating Range	**3.20 – 5.00**
Highest Frequency Area	**Around 4.20–4.30**
Lowest Ratings	**Around 3.20–3.40**

Overall Rating Level	**Good / Positive**

Distribution Trend : -

The histogram shows that most customer ratings are concentrated **between approximately 4.0 and 4.6.**

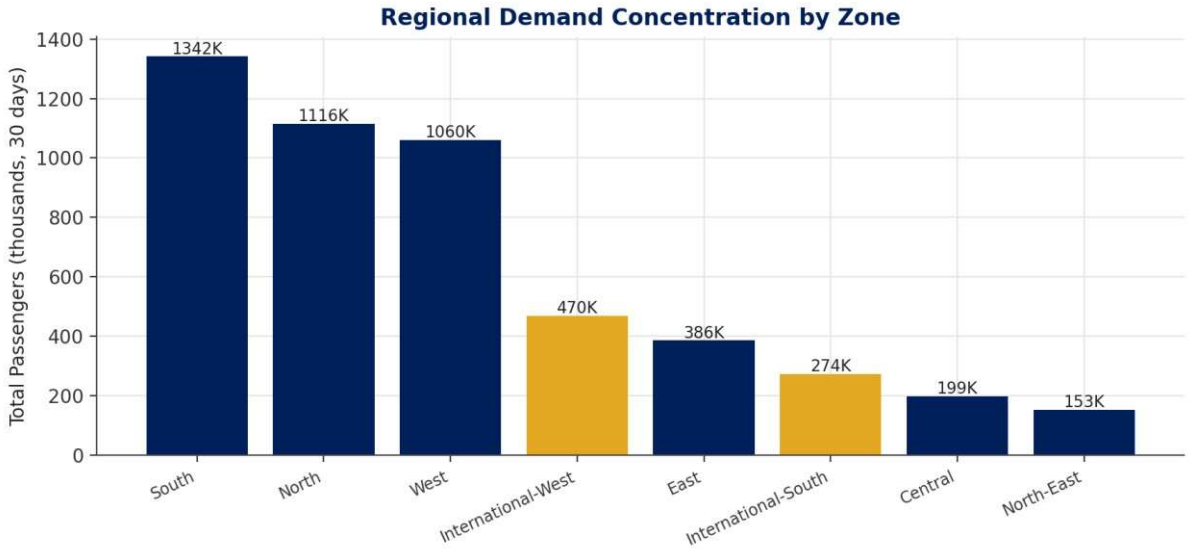
- Ratings increase strongly from **3.5 to 4.0.**
- The largest concentration is around **4.2–4.3.**
- After **approximately 4.3**, the frequency gradually decreases.
- **Very high ratings close to 5.0** are less frequent.
- **Very low ratings below 3.5** are also relatively uncommon**.**

Key Highlight : -

IndiGo's overall customer rating is positive, with an average rating of 4.19/5. Most ratings are concentrated above 4.0, indicating generally good customer satisfaction.

6.Regional Demand :-

Demand is meaningfully concentrated: the South, North, and West zones together account for roughly 71% of total passengers (South ~1.34M, North ~1.12M, West ~1.06M), consistent with IndiGo's real-world hub structure around Delhi, Mumbai, and Bengaluru. International sectors to the Gulf and Southeast Asia (~750K combined) trail domestic trunk demand but carry a materially higher average fare. This view is what drives.



Total passengers by demand zone over the 30-day window.

API key :-

Region	Passengers
South	**1,342K**
North	**1,116K**
West	**1,060K**
International-West	**470K**
East	**386K**
International-South	**274K**
Central	**199K**
North-East	**153K**

Regional Share : -

Region	Share of Total

South	**26.84%**
North	**22.32%**
West	**21.20%**
International-West	**9.40%**
East	**7.72%**
International-South	**5.48%**
Central	**3.98%**
North-East	**3.06%**

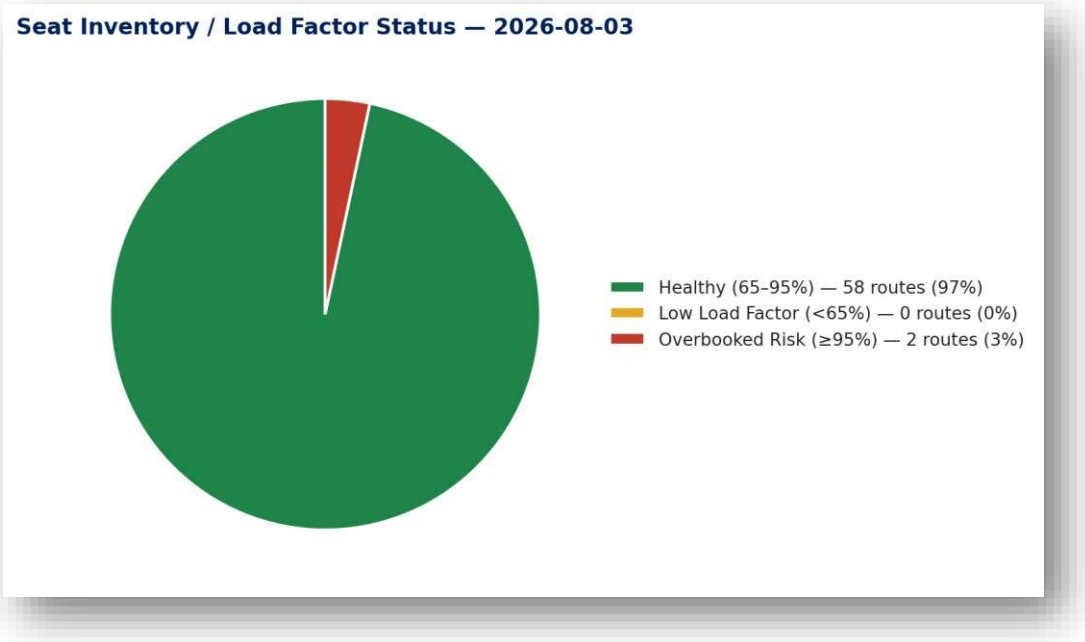
Key Highlights :-

- **South is the strongest demand region**, contributing **26.84%** of total passengers.
- **North and West** are the next two major markets, with **22.32%** and **21.20%** respectively.
- The **South + North + West** regions together account for **70.36%** of total passenger demand.
- **International-West** is the strongest international region with **470K passengers**.
- **North-East has the lowest demand**, with only **153K passengers**.
- Domestic regions dominate the passenger volume compared with international regions.
- The large concentration in South, North, and West suggests that IndiGo should prioritize **flight capacity, frequency, and operational resources** in these high-demand regions.

7.Seat Inventory / Load Factor Status :-

Each route-day was bucketed into three load-factor tiers (passengers ÷ available seats):

- Healthy **(65–95% load factor)** — comfortable buffer, normal revenue management
- **Low Load Factor (< 65%)** — under-performing capacity; candidate for fare stimulation or capacity reallocation
- Overbooked Risk **(≥ 95% load factor)** — bump/denied-boarding risk, should be crossreferenced against the top-route list above; **a hero route flying at 95%+ load** is a materially bigger revenue-management problem than a thin route doing the same
- Most actionable view: “days of supply remaining” per route, cross-tabbed against booking velocity.



Route-day count by load-factor tier, most recent day (60 routes).

The most actionable version of this chart in production cross-tabs load-factor tier against route-level revenue contribution, producing a "seats at risk / seats left to sell" metric per flight rather than a static load-factor snapshot.

KPI Summary :-

KPI	Value

Total Routes	**60**
Healthy Load Factor (65–95%)	**58 routes (97%)**
Low Load Factor (<65%)	**0 routes (0%)**
Overbooked Risk (≥95%)	**2 routes (3%)**
Healthy + Low + Risk	**60 routes**

Key Analysis :-

- **58 out of 60 routes (97%) are in the healthy range**, indicating strong and balanced seat utilization.
- **No route has a low load factor below 65%**, suggesting there is very little unused seat capacity.
- **2 routes (3%) are at overbooking risk**, with load factors of **95% or higher**.
- The very large green section shows that **overall seat inventory is being utilized efficiently**.
- The absence of low-load routes indicates **strong passenger demand across the network**

Summary of Key Metrics :-

Metric	Value (simulated)
Total passengers (30 days)	~4.99M

Total revenue (30 days)	~₹2,961 Cr
Average customer rating	4.19 / 5
Top single-day route	DEL-BOM — 17,503 passengers
Demand zones tracked	8
Routes at Overbooked Risk or Low Load	2 of 60

Table 2. Headline metrics for the 30-day reporting window.

1.Passenger Performance

- IndiGo handled approximately **4.99 million passengers** in 30 days.
- Average daily passenger volume was approximately **1.66 lakh**.
- Passenger demand is concentrated particularly in major regions such as **South, North, and West**.

2. Revenue Performance

- Total simulated revenue was approximately **₹2,961 crore**.
- Average daily revenue was approximately **₹98.7 crore**.
- Estimated revenue per passenger is approximately **₹5,934**.

3. Route Performance

- **DEL–BOM** was the highest-volume route with **17,503 passengers**.
- Delhi and Mumbai appear prominently among the high-demand routes.
- High-demand routes should receive appropriate capacity and frequency planning.

4. Customer Satisfaction

- Average customer rating = **4.19/5**.
- This indicates an overall **positive customer experience**.
- However, some routes showed declining 7-day rolling ratings and should be monitored.

5.Operational Risk

- *Only **2 out of 60 routes (3.33%)** are classified as having overbooked/low-load risk.*
- ***58 routes (96.67%)** remain in the healthy load-factor range.*
- *This indicates generally **efficient seat utilization**.*

Next Steps: From Descriptive → Predictive

1.Passenger Demand Forecasting :-

Use historical passenger data to predict future passenger volume.

Example:-

Predict the number of passengers IndiGo may carry on each route over the next 7, 30, or 90 days.

Possible methods**: Time Series, Linear Regression, Random Forest.**

2. Revenue Prediction

Build a model to estimate future revenue using factors such as:

- Passenger volume
- Ticket price
- Route
- Flight frequency
- Load factor

- Season/month

Output: **Expected daily or monthly revenue.**

3. Load Factor Prediction : -

Predict whether a route will have:

- Healthy load: **65–95%**
- Overbooking risk**: ≥95%**
- Low load**: <65%**

This can help IndiGo optimize seat capacity and flight frequency.

4. Customer Rating Prediction : -

Use historical ratings and operational factors such as:

- Flight delays
- Cancellations
- Route
- Passenger volume
- Service performance to predict future customer satisfaction.

6. Route Risk Prediction ;-

Identify routes that are likely to experience:

- Falling customer ratings
- Low passenger demand
- Overbooking
- Revenue decline
- A risk score can be created for each route.

THANK YOU